

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458375.90048 +/- 0.00064 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1353 +/- 0.0076
Transit depth (Rp/Rs)^2	1.83 +/- 0.21 [%]
Orbital Inclination (inc)	86.82 +/- 0.96
Airmass coefficient 1 (a1)	1.236 +/- 0.011
Airmass coefficient 2 (a2)	-0.1518 +/- 0.007
Scatter in the residuals of the lightcurve fit is	0.9 %
Best Comparison Star	None
Optimal Aperture	4.66
Optimal Annulus	9.69
Transit Duration (day)	0.0867 +/- 0.0043

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1353 $\pm$ 0.0076 vs 0.1404 (deviation 0.67 σ)
Duration fit vs published	0.0867 d vs 0.0754 d (deviation 2.61 σ)
Mid-time O-C	-1.5 min
Depth SNR	17.7
Residual scatter	0.9 %

Light curve

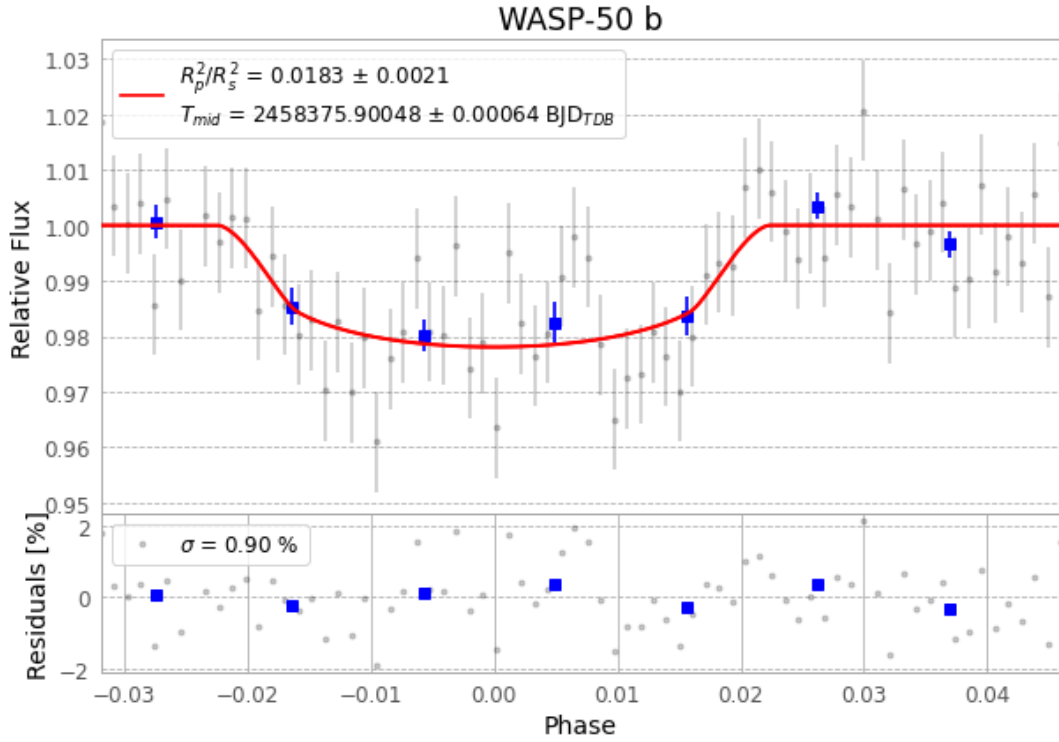


Figure 1 — FinalLightCurve_WASP-50 b_2018-09-14.png (SHA-256 b91b51113b1b074c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [307, 211], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 506bbfbb163ce971...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

74 FITS frames (49 MB), WASP-50180914080012.FITS → WASP-50180914113912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-180914.log (2d0d0a4313c7...), FinalLightCurve_WASP-50 b_2018-09-14.png (b91b51113b1b...), FinalLightCurve_WASP-50 b_2018-09-14.csv (3677bb5aafbc...)

Compute resources

Wall time 115.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 05:30Z.